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|| Jai Sri Gurudev ||
Sri Adichunchanagiri Shikshana Trust (R.)

SJC INSTITUTE OF TECHNOLOGY

(An Autonomous Institution under VTU from 2024-25)

AICTE Approved, Accredited by NBA (CSE, ISE, ECE, ME, CV, AE) & NAAC with A+ Grade, QS I-Gauge Gold rated
Chikkaballapura - 562 101, Karnataka



Estd. 1986



AVINYA 2.0

24-HOUR HACKATHON

PROBLEM STATEMENTS

Theme-1: Artificial Intelligence for Social Good

A. AI-Powered Healthcare diagnostics

1. AI-based medical imaging (X-ray, CT, MRI analysis)
2. Predictive analytics for disease outbreaks
3. Wearable health monitoring systems
4. AI chatbots for primary healthcare support

B. Smart agriculture using ML

1. Crop Disease Detection using Image Processing
2. Soil Quality and Moisture Prediction
3. Smart Irrigation Systems
4. Yield Prediction Models

C. Disaster prediction & response systems

1. Flood, Earthquake, and Cyclone Prediction Models
2. Real-time Disaster Monitoring Systems
3. AI-based Rescue Route Optimization
4. Social Media Data Analysis for Emergency Alerts

D. Personalized Education Platforms

1. Adaptive Learning Systems Based on Student Performance
2. AI Tutors and Chatbot-Based Learning Assistants
3. Learning Analytics and Performance Prediction
4. Content Recommendation Systems

Theme-2: Sustainable Mobility for the Future

A. Hypersonic Vehicle Aerodynamic Design

1. Computational Fluid Dynamics (CFD) for High-Speed Flows
2. Thermal Protection Systems (TPS)

B. Navigate, Adapt & Survive in Adverse Flight Conditions

1. Adaptive Flight Trajectory Management under Dynamic Weather Constraints
2. Intelligent Turbulence and Icing Awareness System for Sustainable Flight Operations